

INTRO

Hoya Botto

Automation for all Kesennuma manufacturing.

One flap is the first test. A local cooperative is the ambition.

Ron Turetzky · Hackatsuon 2026



TEAM

A team for a local manufacturing automation program



Ron Turetzky

Community builder
Bread Cooperative co-founder
Second Kesennuma Hackatsuon



Decentral Park

Selected organizations in Bread's published portfolio



Stellar



EigenCloud



Funding the Commons



AI and software integration
Interface design and workshops
Manufacturing program lead



FIELDWORK

Hachiyo: our first manufacturing case

A food-packaging visit reveals work between machines and people



株式会社 **ハチヨウ**



Automatic filling



People inspect and close cartons

On 2 October, we visited Hachiyo to understand the packaging work.
Its public account shows automation handing work back to people for checking and closing.
First test: one flap. Then assess handling, inspection and other work across manufacturing.





03 FOLD AND CLOSE

Close the top carton's flaps

3D illustration from fieldwork. Timing and layout simplified.

Three-carton sequence: team fieldwork, 2 Oct 2026; private recording retained. Original 3D illustration; timing simplified, no robot result.

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Public process context

The whole manufacturing sector

151 establishments across food, materials, machinery and more

151

manufacturing establishments

78 food + 73 other manufacturing

3,816

people employed

Across all manufacturing categories

¥11.84bn

annual personnel cost

Published 2024 cost base

株式会社 **ハチヨウ**

Food processing

Automatic filling and packing;
people inspect and close cartons.

 **シマ精工**
SHIMASEIKO

Precision manufacturing

Machining and production equipment;
assess loading and inspection.



オフセット印刷・ラベル印刷
双葉印刷株式会社

Printing and finishing

Printing, folding and binding machines;
assess the handling between steps.

Assess what remains manual around existing equipment, sector by sector.



DEMO

First test: one box flap

A small test within a manufacturing-wide program

DEMO PLACEHOLDER

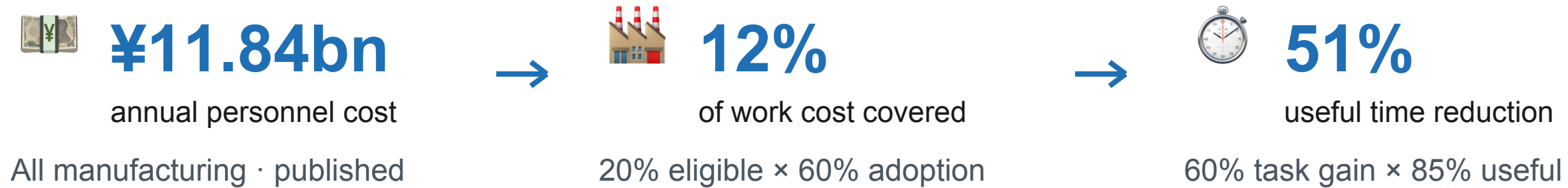
One box · one flap

Live demonstration or verified recording

ECONOMICS

Potential annual value across manufacturing

Published labor-cost base × work reached × useful time released



ILLUSTRATIVE ANNUAL VALUE · BEFORE CAPITAL AND RUNNING COSTS

¥725m

¥11.84bn × 12% × 51%

Broader uptake + task coverage

¥1.79bn

30% × 80% × 70% × 90%



BEYOND BOXES

Beyond boxes, across manufacturing

Every published manufacturing category is in scope; task fit is assessed locally

 Food 78 Food production Inspect · handle · package	 Beverages + feed 10 Beverages, tobacco, feed Fill · label · palletize	 Textiles 7 Clothing + knitwear Inspect · cut · handle	 Materials + print 24 Wood, paper, plastics + more Feed · finish · check quality
 Metals 7 Metal products + nonferrous Load · finish · inspect	 Machinery + electrical 15 Machines + electronics Assemble · test · tend machines	 Transport equipment 8 Shipbuilding + related Fabricate · handle · inspect	 Other manufacturing 2 Other products Choose tasks with each factory

151 establishments

Different processes. Reusable task families.

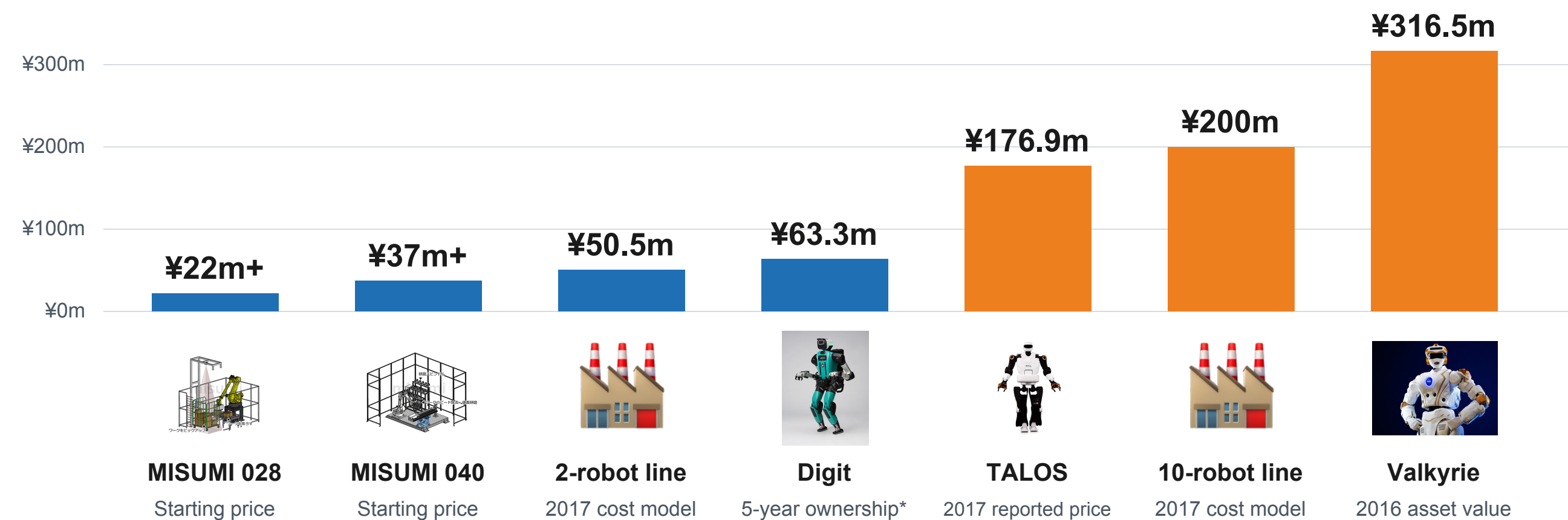
Feeding · handling · inspection · assembly · finishing · packaging

e-Stat 2025 survey, municipal reference table, rows 1498–1518. 21 reported manufacturing categories grouped into 8; task suggestions are ours.



Industrial automation needs serious capital

Published system and humanoid references · every amount shown in yen



Different scopes / dates. *Digit: illustrative five-year ownership. Historical prices / values are not current quotes. Photos: MISUMI, Agility, PAL, NASA. 09 / 14

SHARED DEVELOPMENT

Share automation across manufacturing

Co-ops already work here. Share designs, learning and support across different tasks.

気仙沼鹿折加工協同組合

Co-ops already work here.

Shared facilities. Shared people.

Now share automation.

151 manufacturing sites · 6 reusable task families



ONE CELL / SITE · PLANNING MODEL



Separate: 151 × ¥50m. Shared: 151 × ¥40m + 6 reusable cores × ¥50m.

ONE-TIME DIFFERENCE

¥1.21bn

16% less total cost

Cost model: our unquoted assumptions; not measured savings. Company marks: official sites (full links in notes). Logos do not imply membership.

CITY ALIGNMENT

Built around Kesennuma's priorities

Manufacturing productivity, local enterprise and relationships that continue



生産性向上

Higher productivity

Improve work across manufacturing



新産業・新事業

Local businesses + skills

Train, build and test new services



関係人口

Lasting connections

Return, teach and work together

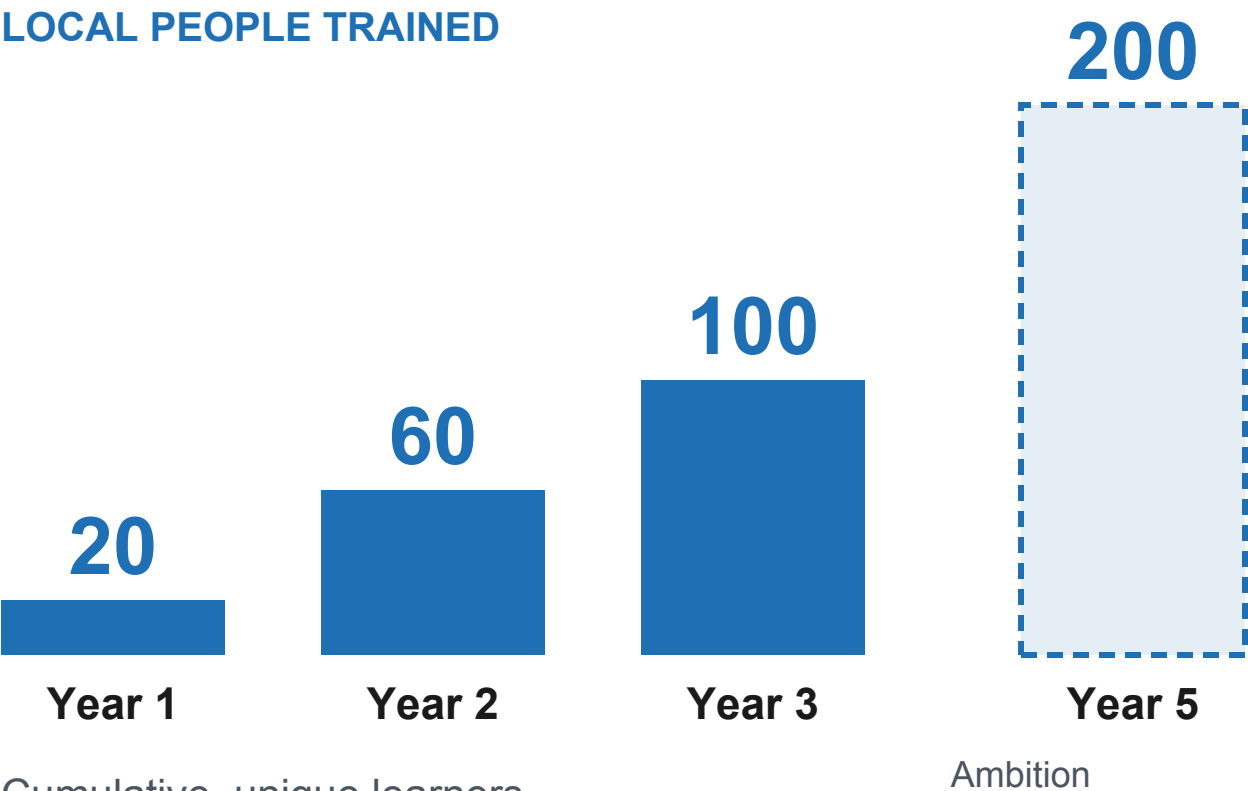


CITY OUTCOMES

Manufacturing skills and local businesses

Operators, technicians and new ventures across the sector

LOCAL PEOPLE TRAINED



Cumulative, unique learners

- 10 Local trainers by year 3**
Graduates teach future cohorts.
Operators, technicians and young makers.
- 20 Factories across manufacturing**
Food, materials, machinery and more.
40 task trials; production needs a quote.
- 5 Local venture pilots by year 3**
Test products and paid services with customers.
Potential new businesses and local inventions.

Years 4–5: further cohorts, local ownership and new businesses if customer demand supports them.

All training, trial and venture numbers: our proposed targets, not published outcomes. Sources below support policy context / team capability only.



WHY NOW

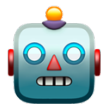
Global robotics is accelerating

AI makes more local manufacturing capability possible. Use the global momentum.

GLOBAL INDUSTRY IS MOVING

+11%

Industrial robot installations
worldwide in 2025
600,000+ new robots



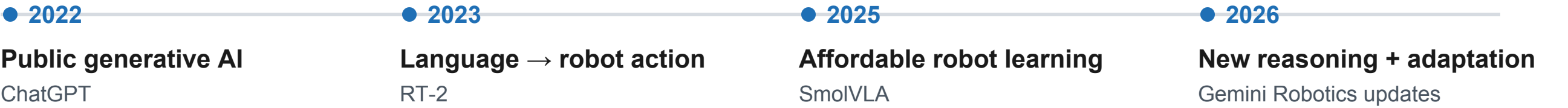
Now possible

Open robot AI can run
on consumer hardware.
More is within small teams' reach.



Use the momentum

Build and learn locally.
Share progress across factories.
Grow skills and new businesses.



Global momentum makes local capability possible.

Primary sources below. Global installations are not local AI adoption. Selected milestones; factory reliability still needs testing.

ROADMAP

Fund a manufacturing-wide program

A ¥11m pilot, then a ¥300m development and training program if it succeeds

90 DAYS



¥11m

Pilot + handover

One flap; measure the value
Train 1 lead + 2 operators

YEAR 1



¥60m

Try other processes

20 learners · 3 sites
6 trials across task families

YEAR 2



¥100m

Reuse across sectors

60 learners · 10 sites
20 cumulative task trials

YEAR 3



¥140m

Local services + ventures

100 learners · 20 sites
40 trials · 5 venture pilots

¥300m follow-on program

Development, trials, training and venture support

¥100m/year savings target

40 production tasks × ¥2.5m net annual saving

Production equipment needs separate site funding. Target savings depend on conversion to production.

Year 5 ambition: 200 learners and locally run services; further funding required.

Our proposed budgets / targets; unquoted and conditional. Production capex is additional. Bread source describes capability, not funding or savings.

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Bread · team capability